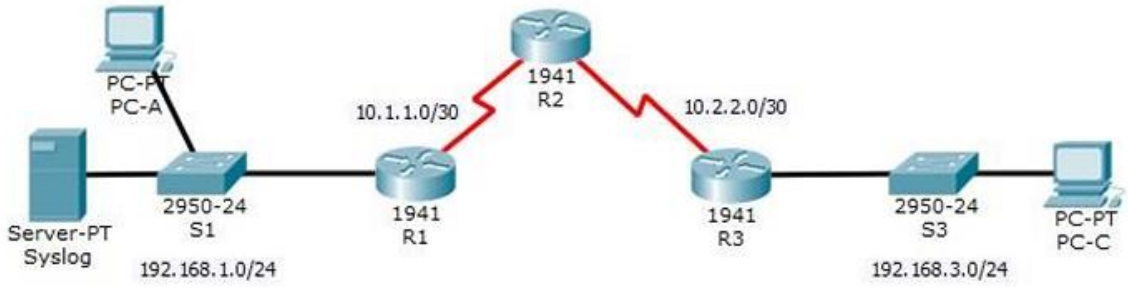
Practical 7: Configure IOS Intrusion Prevention System (IPS) Using the CLI



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

Objectives

• Enable IOS IPS.

• Configure logging.

• Modify an IPS signature.

• Verify IPS.

Routing

Router 1:

Router(config)#hostname R1

R1(config)#router rip

R1(config-router)#version 2

R1(config-router)#network 192.168.1.0

R1(config-router)#network 10.1.1.0

Router 2:

Router(config)#hostname R2

R2(config)#router rip

R2(config-router)#version 2

R2(config-router)#network 10.1.1.0

R2(config-router)#network 10.2.2.0

Router 3:

Router(config)#hostname R3

R3(config)#router rip

R3(config-router)#version 2

R3(config-router)#network 10.2.2.0

R3(config-router)#network 192.168.3.0

Part 1: Enable IOS IPS

Step 1: Enable the Security Technology package.

a. On R1, issue the show version command to view the Technology Package license information.

R1#show version

```
License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX1524P70E


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      None          None          None
data          None          None          None

Configuration register is 0x2102
```

b. If the Security Technology package has not been enabled, use the following command to enable the package.

R1(config)#license boot module c1900 technology-package securityk9

c. Accept the end-user license agreement.

d. Save the running-config and reload the router to enable the security license.

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#reload
```

Press enter for destination filename

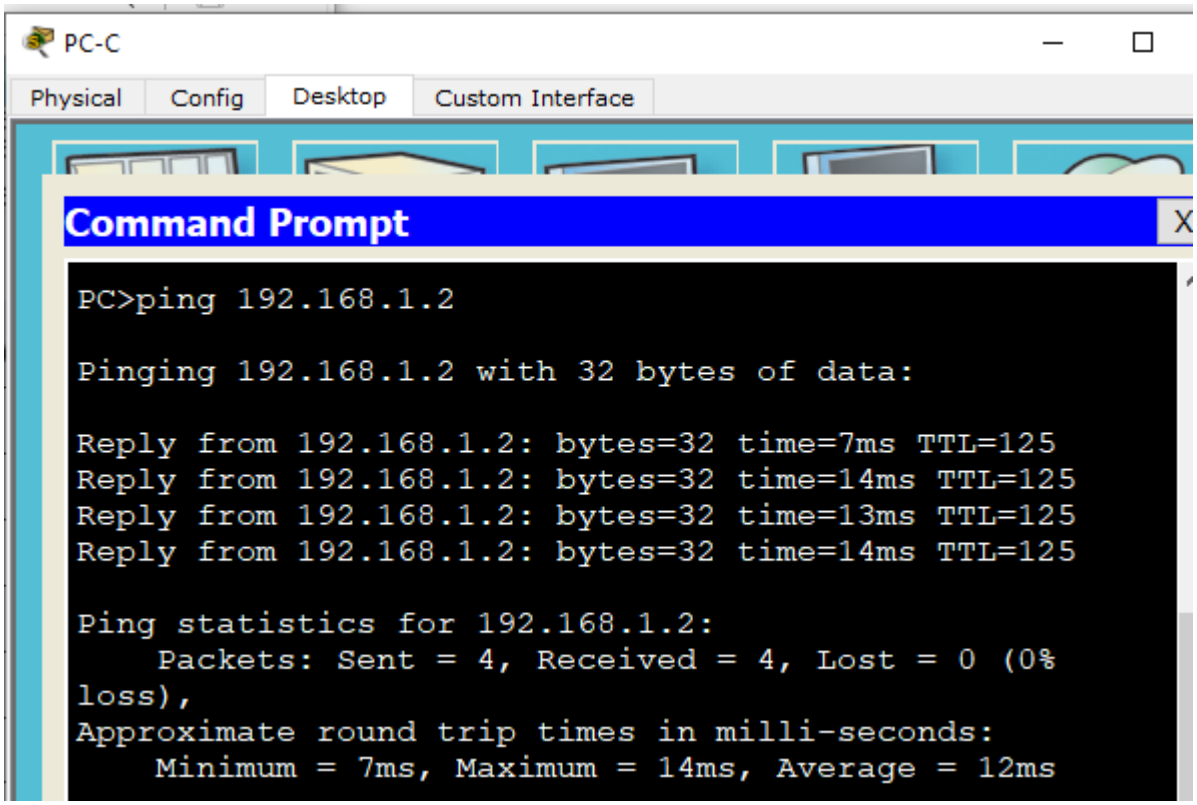
e. Verify that the Security Technology package has been enabled by using the show version command.

R1#show version

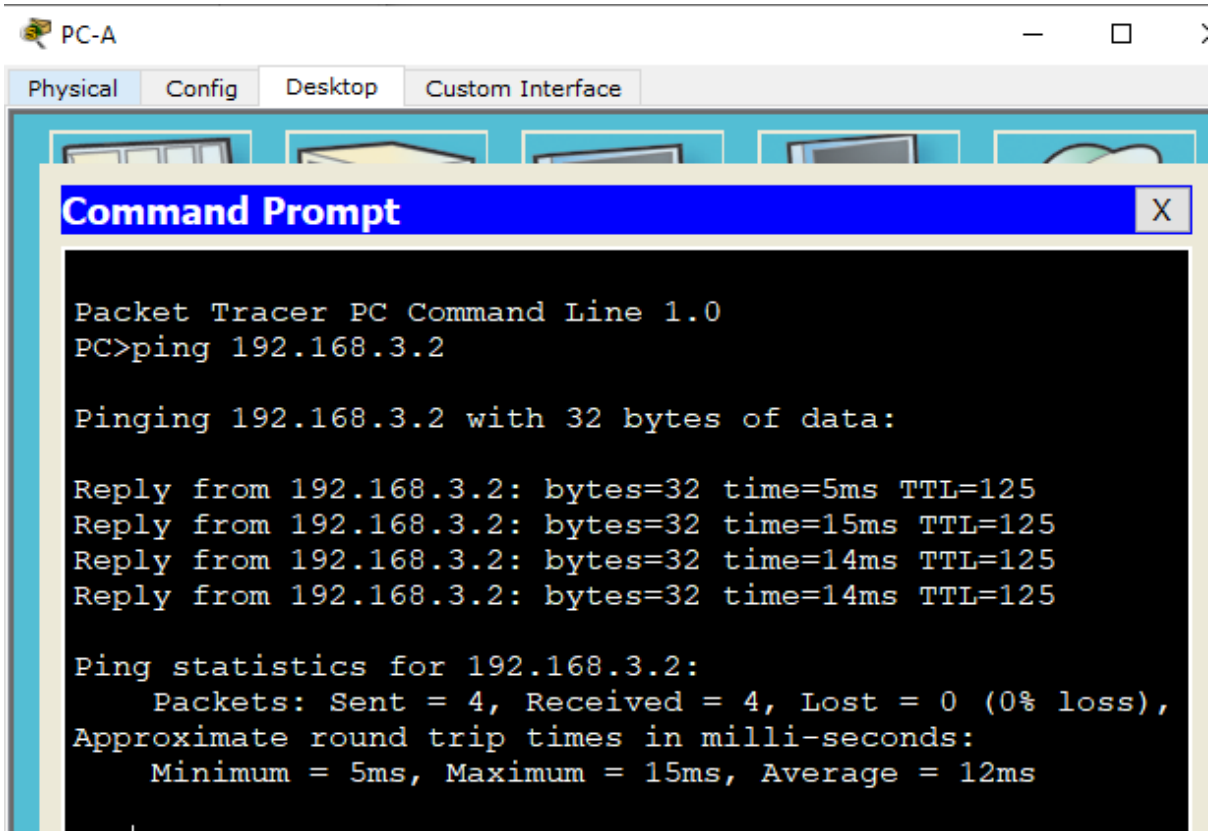
```
License Info:

License UDI:

-------------------------------------------------
Device#    PID                    SN
-------------------------------------------------
*0         CISCO1941/K9           FTX1524P70E


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          None          None          None

Configuration register is 0x2102
```

Step 2: Verify network connectivity.

a. Ping from PC-C to PC-A. The ping should be successful.



b. Ping from PC-A to PC-C. The ping should be successful.



Step 3: Create an IOS IPS configuration directory in flash.

```
R1#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir
```

Step 4: Configure the IPS signature storage location.

R1(config)#ip ips config location flash:ipsdir

Step 5: Create an IPS rule.

R1(config)#ip ips name iosips

Step 6: Enable logging.

R1(config)# ip ips notify log

R1#clock set 16:20:10 8 feb 2026

R1(config)#service timestamps log datetime msec

R1(config)#logging host 192.168.1.50

Step 7: Configure IOS IPS to use the signature categories.

R1(config)#ip ips signature-category

R1(config-ips-category)#category all

R1(config-ips-category-action)#retired true

R1(config-ips-category-action)#exit

R1(config-ips-category)#category ios_ips basic

R1(config-ips-category-action)#retired false

R1(config-ips-category-action)#exit

R1(config-ips-category)#exit

Do you want to accept these changes? [confirm] <enter>

Step 8: Apply the IPS rule to an interface.

R1(config)#interface gigabitEthernet 0/1

R1(config-if)#ip ips iosips out

Part 2: Modify the Signature

Step 1: Change the event-action of a signature.

R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#enabled true

R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action deny-packet-inline
R1(config-sigdef-sig-engine)#exit
R1(config-sigdef-sig)#exit
R1(config-sigdef)#exit
Do you want to accept these changes? [confirm] <enter>

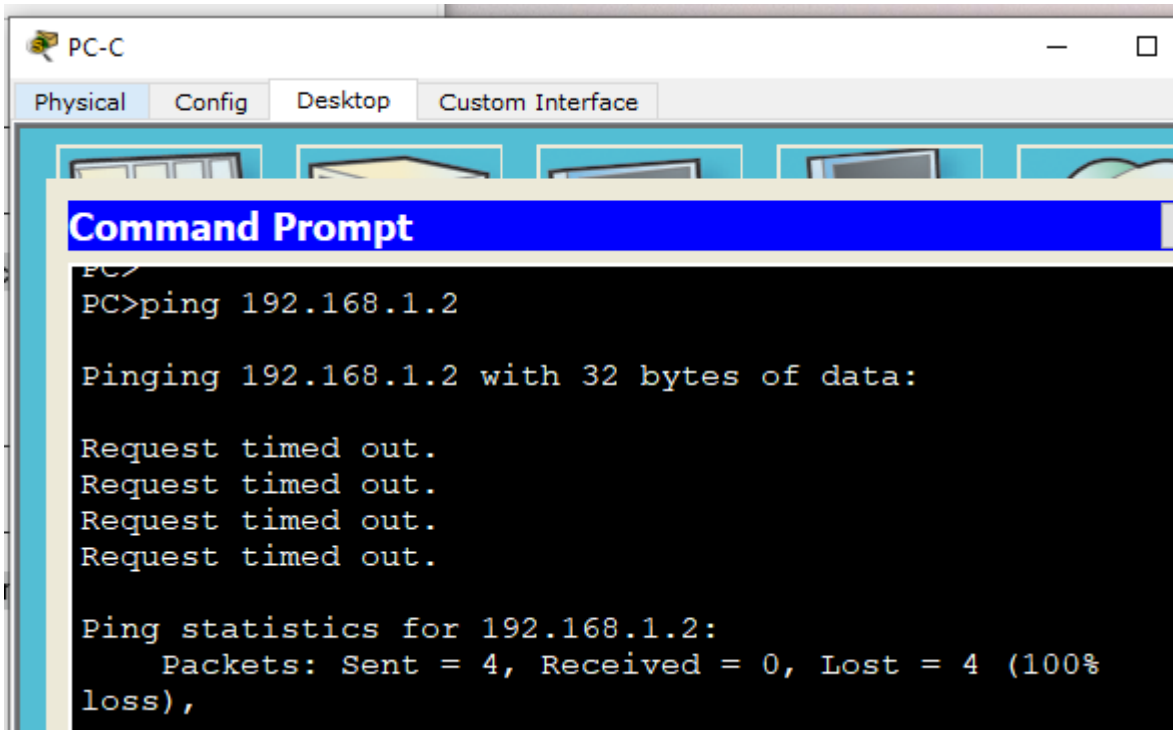
Step 2: Use show commands to verify IPS.

R1#show ip ips all

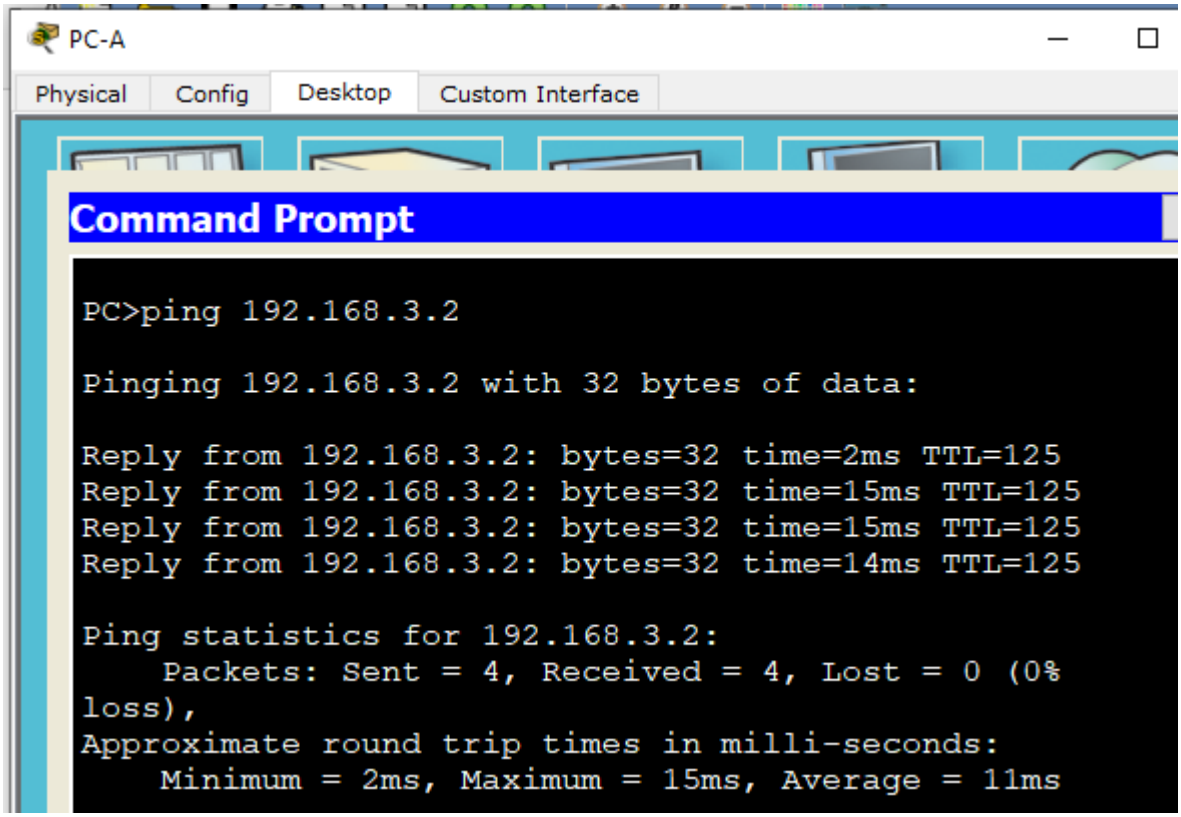
```
IPS Packet Scanning and Interface Status
    IPS Rule Configuration
      IPS name iosips
    IPS fail closed is disabled
    IPS deny-action ips-interface is false
    Fastpath ips is enabled
    Quick run mode is enabled
    Interface Configuration
      Interface GigabitEthernet0/1
        Inbound IPS rule is not set
        Outgoing IPS rule is iosips
```

Step 3: Verify that IPS is working properly.

a. From PC-C, attempt to ping PC-A.



b. From PC-A, attempt to ping PC-C

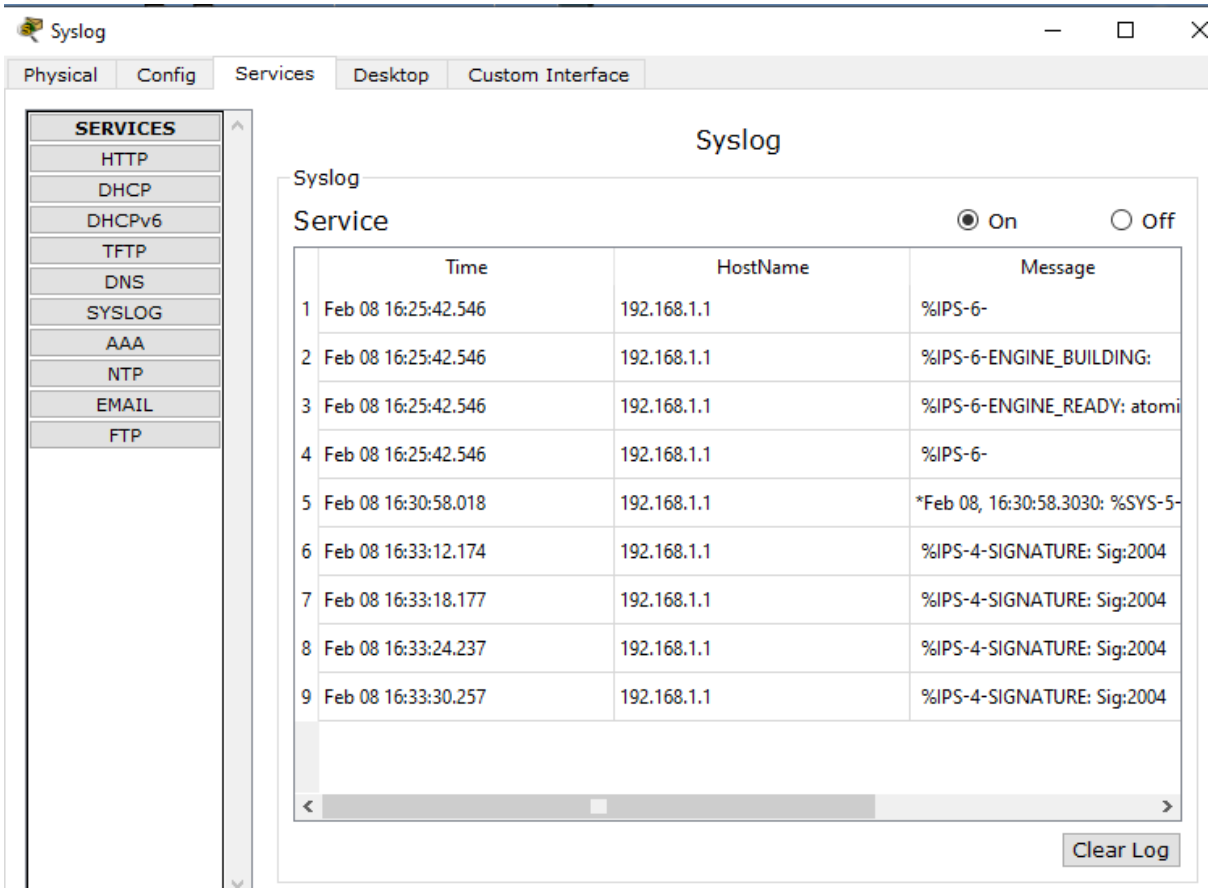


Step 4: View the syslog messages.

a. Click the Syslog server.

b. Select the Services tab.

c. In the left navigation menu, select SYSLOG to view the log file.


Syslog
Physical
Config
Services
Desktop
Custom Interface
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
Syslog
Syslog
Service
On
Off
Time
HostName
Message
1 Feb 08 16:25:42.546 192.168.1.1 %IPS-6-
2 Feb 08 16:25:42.546 192.168.1.1 %IPS-6-ENGINE_BUILDING:
3 Feb 08 16:25:42.546 192.168.1.1 %IPS-6-ENGINE_READY: atomi
4 Feb 08 16:25:42.546 192.168.1.1 %IPS-6-
5 Feb 08 16:30:58.018 192.168.1.1 *Feb 08, 16:30:58.3030: %SYS-5-
6 Feb 08 16:33:12.174 192.168.1.1 %IPS-4-SIGNATURE: Sig:2004
7 Feb 08 16:33:18.177 192.168.1.1 %IPS-4-SIGNATURE: Sig:2004
8 Feb 08 16:33:24.237 192.168.1.1 %IPS-4-SIGNATURE: Sig:2004
9 Feb 08 16:33:30.257 192.168.1.1 %IPS-4-SIGNATURE: Sig:2004
Clear Log